

# GraphTheorySymbol Memo

## v1.2.0 2026/09/06

Add in your preamble: `\usepackage{graph-theory-symbol}`.

| Long name                                              | Short name              | Result  | In context  |
|--------------------------------------------------------|-------------------------|---------|-------------|
| <code>\GTSadjacency/</code>                            | <code>\GTSa/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSadjacencyRelation/</code>                    | <code>\GTSaR/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacency/</code>                          | <code>\GTSu/</code>     | $\circ$ | $u \circ v$ |
| <code>\GTSunadjacencyRelation/</code>                  | <code>\GTSuR/</code>    | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUp/</code>                  | <code>\GTSdAU/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyUpRelation/</code>          | <code>\GTSdAUR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUp/</code>           | <code>\GTSnDAU/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyUpRelation/</code>   | <code>\GTSnDAUR/</code> | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDown/</code>                | <code>\GTSdAD/</code>   | $\circ$ | $u \circ v$ |
| <code>\GTSdirectedAdjacencyDownRelation/</code>        | <code>\GTSdADR/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDown/</code>         | <code>\GTSnDAD/</code>  | $\circ$ | $u \circ v$ |
| <code>\GTSnegatedDirectedAdjacencyDownRelation/</code> | <code>\GTSnDADR/</code> | $\circ$ | $u \circ v$ |
| <code>\GTSincidence/</code>                            | <code>\GTSi/</code>     | $\circ$ | $v \circ e$ |
| <code>\GTSincidenceRelation/</code>                    | <code>\GTSiR/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidence/</code>                     | <code>\GTSnI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedIncidenceRelation/</code>             | <code>\GTSnIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidence/</code>                         | <code>\GTSoI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSoutIncidenceRelation/</code>                 | <code>\GTSoIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidence/</code>                  | <code>\GTSnOI/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedOutIncidenceRelation/</code>          | <code>\GTSnOIR/</code>  | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidence/</code>                          | <code>\GTSiI/</code>    | $\circ$ | $v \circ e$ |
| <code>\GTSinIncidenceRelation/</code>                  | <code>\GTSiIR/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidence/</code>                   | <code>\GTSnII/</code>   | $\circ$ | $v \circ e$ |
| <code>\GTSnegatedInIncidenceRelation/</code>           | <code>\GTSnIIR/</code>  | $\circ$ | $v \circ e$ |

| Long name                                           | Short name  | Result | In context                       |
|-----------------------------------------------------|-------------|--------|----------------------------------|
| \GTSadjacencyMonochrome/                            | \GTSaM/     |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSadjacencyMonochromeRelation/                    | \GTSaMR/    |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSunadjacencyMonochrome/                          | \GTSuM/     |        | $u \xrightarrow{\text{green}} v$ |
| \GTSunadjacencyMonochromeRelation/                  | \GTSuMR/    |        | $u \xrightarrow{\text{green}} v$ |
| \GTSdirectedAdjacencyUpMonochrome/                  | \GTSdAUM/   |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSdirectedAdjacencyUpMonochromeRelation/          | \GTSdAUMR/  |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyUpMonochrome/           | \GTSnDAUM/  |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyUpMonochromeRelation/   | \GTSnDAUMR/ |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSdirectedAdjacencyDownMonochrome/                | \GTSdADM/   |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSdirectedAdjacencyDownMonochromeRelation/        | \GTSdADMR/  |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyDownMonochrome/         | \GTSnDADM/  |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyDownMonochromeRelation/ | \GTSnDADMR/ |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSincidenceMonochrome/                            | \GTSiM/     |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSincidenceMonochromeRelation/                    | \GTSiMR/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedIncidenceMonochrome/                     | \GTSnIM/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedIncidenceMonochromeRelation/             | \GTSnIMR/   |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSoutIncidenceMonochrome/                         | \GTSoIM/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSoutIncidenceMonochromeRelation/                 | \GTSoIMR/   |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedOutIncidenceMonochrome/                  | \GTSnOIM/   |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedOutIncidenceMonochromeRelation/          | \GTSnOIMR/  |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSinIncidenceMonochrome/                          | \GTSiIM/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSinIncidenceMonochromeRelation/                  | \GTSiIMR/   |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedInIncidenceMonochrome/                   | \GTSnIIM/   |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedInIncidenceMonochromeRelation/           | \GTSnIIMR/  |        | $v \xrightarrow{\text{blue}} e$  |
| <hr/>                                               |             |        |                                  |
| \GTSsetMainColor{blue}                              |             |        |                                  |
| Long name                                           | Short name  | Result | In context                       |
| \GTSadjacency/                                      | \GTSa/      |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSadjacencyRelation/                              | \GTSaR/     |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSunadjacency/                                    | \GTSu/      |        | $u \xrightarrow{\text{green}} v$ |
| \GTSunadjacencyRelation/                            | \GTSuR/     |        | $u \xrightarrow{\text{green}} v$ |
| \GTSdirectedAdjacencyUp/                            | \GTSdAU/    |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSdirectedAdjacencyUpRelation/                    | \GTSdAUR/   |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyUp/                     | \GTSnDAU/   |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyUpRelation/             | \GTSnDAUR/  |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSdirectedAdjacencyDown/                          | \GTSdAD/    |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSdirectedAdjacencyDownRelation/                  | \GTSdADR/   |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyDown/                   | \GTSnDAD/   |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSnegatedDirectedAdjacencyDownRelation/           | \GTSnDADR/  |        | $u \xrightarrow{\text{blue}} v$  |
| \GTSincidence/                                      | \GTSi/      |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSincidenceRelation/                              | \GTSiR/     |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedIncidence/                               | \GTSnI/     |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedIncidenceRelation/                       | \GTSnIR/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSoutIncidence/                                   | \GTSoI/     |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSoutIncidenceRelation/                           | \GTSoIR/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedOutIncidence/                            | \GTSnOI/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedOutIncidenceRelation/                    | \GTSnOIR/   |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSinIncidence/                                    | \GTSiI/     |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSinIncidenceRelation/                            | \GTSiIR/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedInIncidence/                             | \GTSnII/    |        | $v \xrightarrow{\text{blue}} e$  |
| \GTSnegatedInIncidenceRelation/                     | \GTSnIIR/   |        | $v \xrightarrow{\text{blue}} e$  |